



Natural Language Processing

Part of Speech

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Natural Language Processing

Parts of Speech

In this lecture:



Part 1: **POS Overview**

Part 2: POS Tagging (Viterbi algorithm)

Part 3: ALMA Morph Tagger

Part 4: Discussion

What is Parts of Speech (POS)

Based on [1]

In traditional grammar, a part of speech (POS) is a category of words, or, more generally, of lexical items, that **have similar grammatical properties**.

Words that are assigned to the same part of speech generally display **similar syntactic and semantic behavior**, and sometimes they **undergo morphological inflection similarly**.

The History of Parts of Speech

The first description of parts of speech could be attributed to Dionysius Thrax of Alexandria (c. 100 B.C.), who described 8 parts of speech **for Greek: noun, verb, pronoun, preposition, adverb, conjunction, participle, and article.**

These POS tags became the basis for descriptions of European Languages.

Parts of Speech in Arabic

Traditional Arabic grammar includes three parts of speech:

noun (اسم), **verb** (فعل), **Functional word** (كلمة وظيفية/حرف).

Computationally, different annotation systems proposed different POS tagsets, inspired by the need to describe lexical units with varying granularities.

The decision of which POS tag set to select depends on the desired task.

Existing POS Tagsets in Arabic

1. Buckwalter tagset: contains around 500 unique tags (for stems and affixes), and untokenized (/combinations), this number gets inflated to over 300k unique tags.

Example :

(الجميلة): DET+ADJ+NSUFF_FEM_SG+CASE_DEF_ACC

This tagset is also used in the [Penn Arabic Treebank](#).

There are [several tagsets that evolved from the Buckwalter tagset](#) with the goal of reducing the number of tagsets to a more manageable number, such as the **Bies** tagset, the **Kulick** tagset, and the **Extended Reduced** tagset.

Existing POS Tagsets in Arabic

1. Buckwalter tagset (Examples of Stem tags)

noun	اسم		
noun_prop	اسم علم	IV	فعل مضارع
adj	صفة	PV	فعل ماضي
abbrev	إختصار	CV	فعل أمر
interj	تعجب	IV_PASS	فعل مضارع مجهول
adv	ظرف	PV_PASS	فعل ماضي مجهول
part	أداة	VERB	فعل
prep	حرف جر	INTERROG_PART+PV	أداة استفهام + فعل ماضي
conj	عطف	PV+SUB_CONJ	فعل ماضي + أداة ربط
adj_comp	صفة مقارنة	NEG_PART+PV	أداة نفي + فعل ماضي
noun_quant	اسم كم		
noun_num	اسم عدد		
adj_num	صفة عدد		

Existing POS Tagsets in Arabic

2. Khoja Tagset:

177 tags: 103 nouns, 57 verbs, 9 particles (أدوات),
7 residuals (زوائد), and 1 punctuation.

Example:

باسمه (by his name) : PPr_NCSgMGI_NPrPSg3M

Existing POS Tagsets in Arabic

3. PADT tagset:

developed to for the **Prague Arabic Treebank**: 20 POS tags, and includes information on 7 features.

Example:

قالوا : VP-A-3MP

Existing POS Tagsets in Arabic

4. CATiB tagset:

developed for the Columbia Arabic Treebank, this tagset was designed with simplicity of annotation in mind, and therefore contains only 6 POS tags.

Example:

المدرسون: NOM

Comparison of Different Arabic POS Tagsets

Based on [2]

Arabic	Gloss	Buckwalter/PATB	CATiB	Bies	Kulick	ERTS	Khoja	PADT	ALMORGEANA/MADA
خمسون xams+uwna	fifty	NOUN_NUM+ NSUFF_MASC_PL_NOM	NOM	CD	CD	CD	NNuCaPiM	QL-----1I	POS:NUM +MASC +PL +NOM
الف Âalf+a	thousand	NOUN_NUM+ CASE_DEF_ACC	NOM	CD	CD	CD	NNuCaSgM	QM-----S4R	POS:NUM +DEF +ACC
سائح sAÿiH+i	tourist	NOUN+ CASE_INDEF_GEN	NOM	NN	NN	NNM	NCSgMGI	N-----S2I	POS:N +INDEF +GEN
زاروا zAr+uwA	visited	PV+ PVSUFF_SUBJ:3MP	VRB	VBD	VBD	VBD	VPPI3M	VP-A-3MP--	POS:V +PV +S:3MS
مدينة madiyn+aĥ+a	city	NOUN+ NSUFF_FEM_SG+ CASE_DEF_ACC	NOM	NN	NN	NNF	NCSgFAI	N-----S4R	POS:N +FEM +SG +DEF +ACC
نا+ +nA	our	POSS_PRON_1P	NOM	PRP\$	PRP\$	PRP\$	NPpPP11	S----1-P2-	+P:1P
الجميلة Al+jamiyl+aĥ+a	beautiful	DET+ADJ+ NSUFF_FEM_SG+ CASE_DEF_ACC	NOM	JJ	DT+JJ	DJJF	NASgFAD	A-----FS4D	POS:AJ Al+ +FEM +SG +DEF +ACC
في fiy	in	PREP	PRT	IN	IN	IN	PPr	P-----	POS:P
أيلول Âay.luwl+a	September	NOUN_PROP+ CASE_INDEF_GEN	PROP	NNP	NNP	NNPM	Rmy	N-----S2I	POS:PN +INDEF +GEN
الماضي Al+mADiy	past	DET+ADJ	NOM	JJ	DT+JJ	DJJM	NASgMGD	A-----MS2D	POS:AJ Al+
.	.	PUNC	PNX	PUNC	.	PUNC	PU	G-----	POS:PX

Existing POS Tagsets in Arabic

5. Qabas tagset:

Developed at SinaLab,
based on SAMA, and used
to tag Qabas and many
corpora

See POS tagset and guidelines
<https://sina.birzeit.edu/qabas/about>

أنواع المدخلات	تصنيف قسم الكلام	
اسم	اسمية	
اسم علم		
صفة		
صفة مقارنة		
صفة عدد		
اسم عدد		
اسم كم		
عدد		
صوت		
حرف-اختصار		
المجموع		
فعل ماضي	فعلية	
فعل مضارع		
فعل أمر		
فعل ماضي مجهول		
فعل مضارع مجهول		
المجموع		
ضمير، ضمير إشارة، اسم موصول، ظرف، أداة موصول، ظرف، أداة استفهام، أداة استثناء، أداة تفصيل، أداة نداء، أداة تعريف، أداة ربط، أداة، حرف جر، حرف جر + أداة تعريف، حرف جر + أداة ربط، حرف جر + أداة نفي، حرف جر + اسم موصول، حرف جر + ضمير استفهام، حرف جر + ضمير إشارة، أداة ربط، أداة نفي، عطف، أداة استقبال، أداة مضارع، أداة نفي، أداة نفي + اسم، فعل ماضٍ + أداة ربط، شبه فعل + اسم موصول، تعجب، جواب شرط + أداة نفي، اسم فعل/ جامد، أداة فعل، شبه فعل، علامة ترقيم، أداة ربط + أداة نفي.		وظيفية

Existing POS Tagsets in Arabic

Functional Words

5. Qabas tagset:

Nominals

اسم	NOUN
اسم علم	NOUN_PROP
صفة	ADJ
صفة مقارنة	ADJ_COMP
صفة عدد	ADJ_NUM
اسم عدد	NOUN_NUM
اسم كم	NOUN_QUANT
عدد	DIGIT
اسم صوت	NOUN_VOICE
حرف-اختصار	ABBREV

Verbs

فعل ماضي	PV
فعل مضارع	IV
فعل أمر	CV
فعل ماضي مجهول	PV_PASS
فعل مضارع مجهول	IV_PASS

ضمير	PRON
ضمير اشارة	DEM_PRON
اسم موصول	REL_PRON
ظرف موصول	REL_ADV
ظرف	ADV
اداة استفهام	INTERROG_PART
ظرف استفهام	INTERROG_ADV
ضمير استفهام	INTERROG_PRON
اداة إستثناء	RESTRIC_PART
اداة تفصيل	FOCUS_PART
اداة نداء	VOC_PART
اداة تعريف	DET
اداة	PART
اداة ربط	SUB_CONJ
حرف جر	PREP
حرف جر + أداة تعريف	PREP + DET
حرف جر + أداة ربط	PREP + SUB_CONJ
حرف جر + أداة نفي	PREP + NEG_PART
حرف جر + اسم موصول	PREP + REL_PRON
حرف جر + ضمير استفهام	PREP + INTERROG_PRON
حرف جر + ضمير اشارة	PREP + DEM_PRON
اداة ربط + أداة نفي	SUB_CONJ + NEG_PART
عطف	CONJ
اداة استقبال	FUT_PART
اداة مضارعة	PROG_PART
اداة نفي	NEG_PART
اداة نفي + اسم	NEG_PART + NOUN
فعل ماضي + أداة ربط	PV + SUB_CONJ
شبه فعل + اسم موصول	PSEUDO_VERB + REL_PRON
تعجب	INTERJ
ضمير تعجب	EXCLAM_PRON
جواب شرط + أداة نفي	RC_PART + NEG_PART
اسم فعل/جامد	VERB
اداة فعل	VERB_PART
شبه فعل	PSEUDO_VERB
علامة ترقيم	PUNC
يموجي	EMOJI
- جواب	

Existing POS Tagsets in Arabic

5. Qabas tagset:

قواعد تصنيف المدخلات حسب قسم الكلام

أولى معجم قيس أهمية كبيرة لتحديد قسم الكلام للمدخلات الواردة فيه، لما لذلك من أهمية كبيرة في التطبيقات الحاسوبية. ولم يكتف المعجم بتقسيم الكلمة على أنها اسم وفعل، بل توسع إلى أكثر من ذلك وخصص لكل منها وسوماً خاصة، فأدرج اسم العلم، واسم الفعل، والأداة، واسم العدد، والعدد، وغيرها من أقسام الكلام. فيما يلي شرح لكل تصنيف وقواعد ضبطه:

أولاً: الاسم

للاسم عشرة وسوم في معجم قيس، وفيما يلي وصف لهذه الوسوم:

اسم

41. يستعمل وسم الاسم) للمدخلات الاسمية التي ليست (اسم علم أو اسم عدد أو اسم كم أو صفة أو صفة مقارنة أو اسم صوت). يحتوي قيس على 36,29,053 مدخلة تم توسيمها باسم. مثل: شُعَيْبِيَّةٌ، مَأْفُوءِيَّةٌ، تَسْطُحٌ، مُرْتَجَبٌ، مُرْتَجَى.

اسم علم

42. لا يستخدم اسم العلم كقسم كلام إلا في حالة واحدة فقط، وهي أن تكون المدخلة لا تحتل إلا معنى اسم العلم. مثل: إِيْلَاش، شِفَاهِي، عَيْشِي، قُوسِي، زُكْرِيَّا، إِشْحَاقُ. يحتوي قيس على 4,319 اسم علم. أما إذا كانت المدخلة تحتل عدة معانٍ وأحد هذه المعاني يفيد معنى اسم العلم فيجب إضافة تعريف يفيد اسم العلم. مثل: مدخلة مُصْطَفَى وبهذه الحالة لا يتم توسيمها باسم علم ولكن إحدى معانيها هو اسم علم.

43. تضاف أسماء الأعلام المشهورة كأسماء المدن والدول والبحار والأنهار وغيرها كمدخلات، ويتم إضافة تعريف دلالي يعرف ويحدد هذه المدخلات/الأعلام - تم استخدام تعريفات من الويكيبيديا في حالة عدم توفر تعريفات معجمية جيدة.

اسم العدد

44. يستعمل للدلالة على الأعداد. يحتوي قيس على 44 اسم عدد، مثل: واحد، اثنان، ثلاث، سِتْعُون، سِتَّةٌ، عَشْرَةٌ، ثَمَانِيَاً | ثَمَانٍ مَائَةٍ.

صفة عدد

45. يستعمل للدلالة على الترتيب، مثل: خَادِي، أَوَّلُ، غَابِرٌ، سَابِعٌ، تَاسِعٌ. يحتوي معجم قيس على 12 مدخلة من نوع صفة عدد.

اسم كم

46. يستعمل للمدخلات التي تدل على الكمية، مثل: بَضْعٌ، بَعْضٌ، كُلٌّ، جَمِيعٌ. يحتوي معجم قيس على 19 اسم كم.

اسم صوت

47. تم إضافة أسماء الأصوات كمدخلات كما وردت في أغلب المعاجم، وإذا اختلفت بعض المعاجم في كتابتها، فيتم ترجيح المعاجم الأكثر شيوعاً، مثل: (حاي) لجزر الإبل، (يس) للقطط، (يس) لتسكين الإبل عند حلبها، (كخ) لجزر الطفل. يحتوي معجم قيس على 16 مدخلة تم توسيمها على أنها اسم صوت.

صفة

48. تستعمل إذا كانت جميع معاني الكلمة تشير إلى (صفة مشبهة، أو نسبة، أو اسم فاعل، أو اسم مفعول، أو تصغير، أو تكبير). ويتم الحكم على هذه الأنواع الستة بناء على الميزان الصرفي. ومثال ذلك: جَلِيٌّ، و بَارِدٌ، و أَيْبِيٌّ التي تشير في جميع معانيها إلى معنى الصفة، لذا يتم توسيمها بصفة. أما مدخلة مُجْعَفٌ فإن أحد معانيها ليس صفة، لذلك تم توسيمها ب(اسم). يحتوي معجم قيس على 11,067 صفة. (انظر قاعدة رقم 9)

صفة مقارنة

49. يستعمل وسم صفة المقارنة مع المدخلات التي على وزن أفعل التفضيل مثل: أعْظَمُ. ويحتوي قيس على 295 مدخلة تم توسيمها بصفة مقارنة.

• عن المعجم

• آخر

• إحصائيات

• مقدمة المعجم

• شمولية معجم قيس

• مبررات تطوير المعجم

• فلسفة المعجم

• المصادر اللغوية

• الدليل المعيارى

• تصنيف المدخلات

• أسئلة شائعة

• حقوق الملكية

• تنزيل قيس

5. Qabas tagset:

- «عن المعجم
- «أخبار
- «إحصائيات
- «مقدمة المعجم
- «شمولية معجم قيس
- «ممرات تطوير المعجم
- «نمسة المعجم
- «المصادر اللغوية
- «الدليل المعياري
- «تصنيف المداخلات
- «أسئلة شائعة
- «حقوق الملكية
- «ترازلي قيس

Natural Language Processing

Parts of Speech

In this lecture:

- ❑ Part 1: POS Overview
- ❑ Part 2: **POS Tagging** (Viterbi algorithm)
- ❑ Part 3: ALMA Morph Tagger
- ❑ Part 4: Discussion

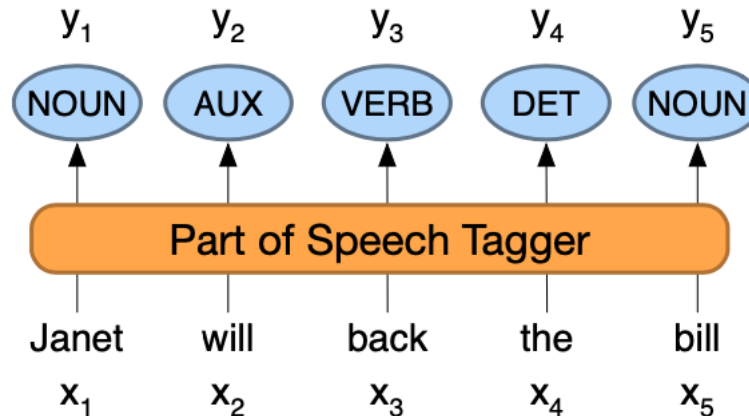


Part-of-Speech Tagging

Based on [3]

Part-of-speech tagging is the process of assigning a part-of-speech to each word in a text.

The input is a sequence $x_1, x_2, \dots, x_n$ of (tokenized) words and a tagset, and the output is a sequence $y_1, y_2, \dots, y_n$ of tags, each output y_i corresponding exactly to one input x_i .



POS Tagging and Ambiguity

Given a word out of context, such as '**back**', POS tagging is a difficult process.

my **back** hurts? (as noun)

I will **back** you up'? (as verb)

Similarly, in Arabic, 'ذهب' could have different tags depending on its contexts:

ذهب الولد إلى المدرسة (as verb)

دفعت ثمنه ذهب (as noun)

POS Tagging and Ambiguity

There are many methods for POS tagging:

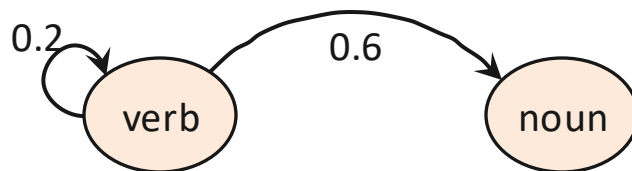
- The Hidden Markov Models using the **Viterbi algorithm** ← Classical Method
- N-grams with SVM
- Frequency-based POS tagging ← ALMA Morph Tagger
- Sequence-to-Sequence using BERT (Transformers)
- Generative Models
-

POS Tagging Using Hidden Markov Model (HMM)

Based on [4]

HMM uses Markov Chains that operate on identifying a word's POS tag based on the POS tag of the previous word.

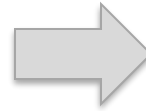
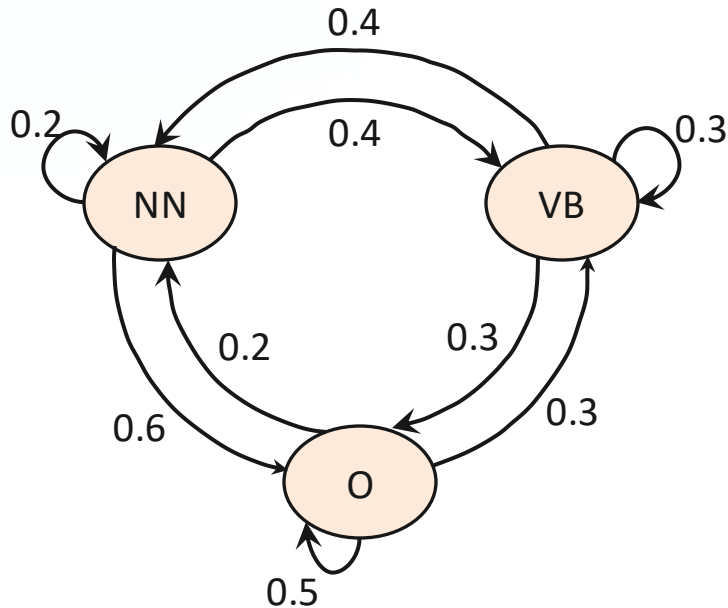
For example, in English, we know that a verb is more likely to be followed by a noun than another verb.



HMM Representation

Based on [4]

HMM are directed graphs, with weights on each edge:



	NN	VB	O
NN(noun)	0.2	0.2	0.6
VB(verb)	0.4	0.3	0.3
O(other)	0.2	0.3	0.5

Building a HMM

The *A matrix* contains the tag transition probabilities $P(t_i | t_{i-1})$ which represent the probability of a tag occurring given the previous tag, using maximum likelihood estimate:

$$P(t_i | t_{i-1}) = \frac{C(t_{i-1}, t_i)}{C(t_{i-1})}$$

Example:

$$P(verb | noun) = \frac{C(noun \ verb)}{C(noun)}$$

These counts are extracted from large manually annotated corpora.

Building a HMM – cont.

The *B matrix* contains the emission probabilities $P(w_i|t_i)$, represent the probability, given a tag, that it will be associated with a given word.

In other words, what is the likelihood of a *verb* tag associating with the word 'back' versus a noun tag?

$$P(w_i|t_i) = \frac{C(t_i, w_i)}{C(t_i)}$$

Example:

$$P(back|verb) = \frac{C(verb\ back)}{C(verb)}$$

Using HMM

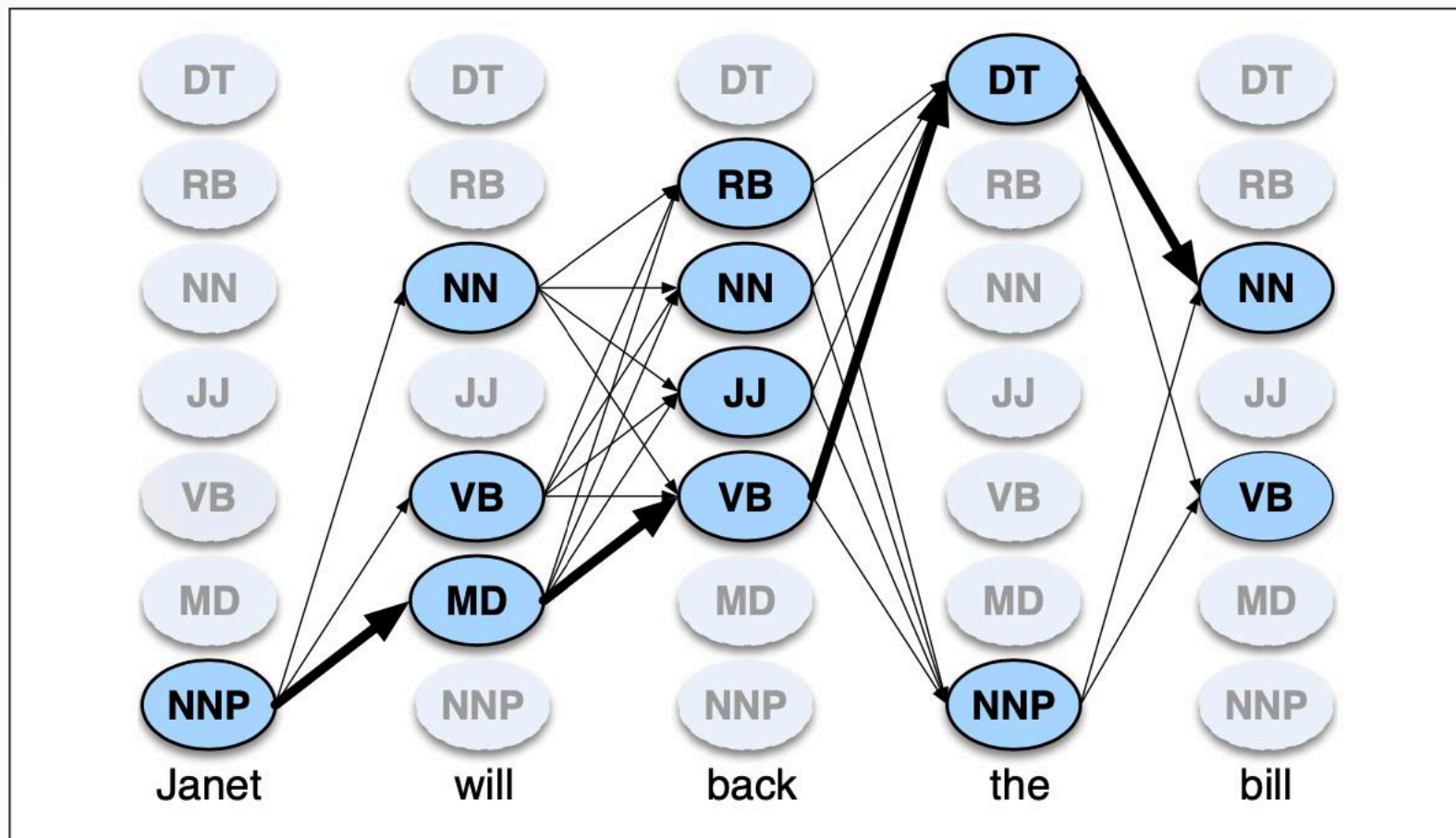
The simplest way to apply the HMM that is represented in matrices A (transition probability) and B (emission probability), is by using a greedy decoder called the **Viterbi algorithm**:

We pick that tag t_i that will maximize the value of:

$$P(t_i|t_{i-1}) * P(w_i|t_i)$$

POS Tagging using HMM with Viterbi Algorithm

Based on [3]



Viterbi Probability Computation

Based on [5]

Given a string s and token t

$$\begin{aligned} \mathbf{viterbi}[s,t] = & \max_{s'} (\\ & \mathbf{veterbi}[s',t-1] \times \\ & \text{transition probability } P(s \mid s') \times \\ & \text{emission probability } P(\text{token}[t] \mid s) \\ &) \end{aligned}$$

For each s,t :

record which s' , $t-1$ contributed the max

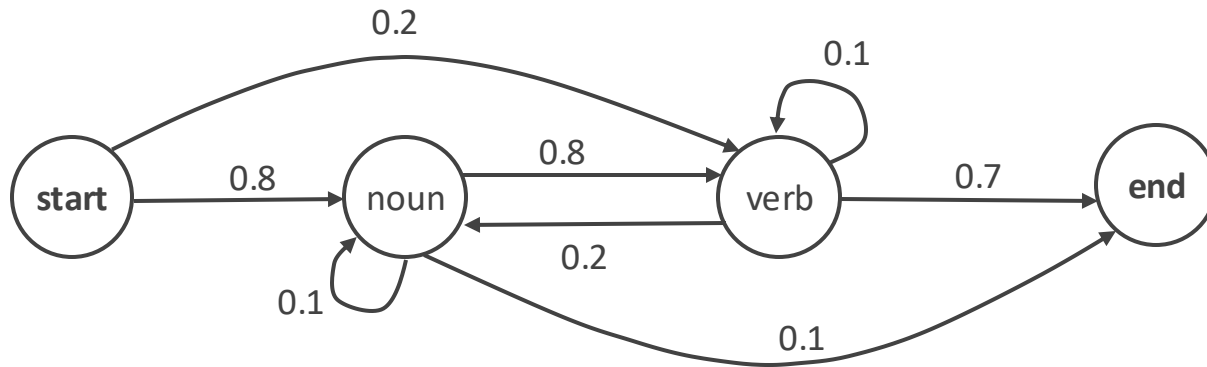
Analysis Example

Based on [5]

Tag the sentence:

Fish sleep.

Using the following simple HMM (transition probabilities):



Word Emission Probabilities

Based on [5]

A two-word language: “*fish*” and “*sleep*”

Suppose in our training corpus:

- “*fish*” appears 8 times as a noun and 5 times as a verb
- “*sleep*” appears 2 times as a noun and 5 times as a verb

Emission probabilities:

Noun

- $P(\text{fish} \mid \text{noun}) : 0.8$
- $P(\text{sleep} \mid \text{noun}) : 0.2$

Verb

- $P(\text{fish} \mid \text{verb}) : 0.5$
- $P(\text{sleep} \mid \text{verb}) : 0.5$


$$\frac{C(\text{noun fish})}{C(\text{noun})} = \frac{8}{2+8} = \frac{8}{10}$$

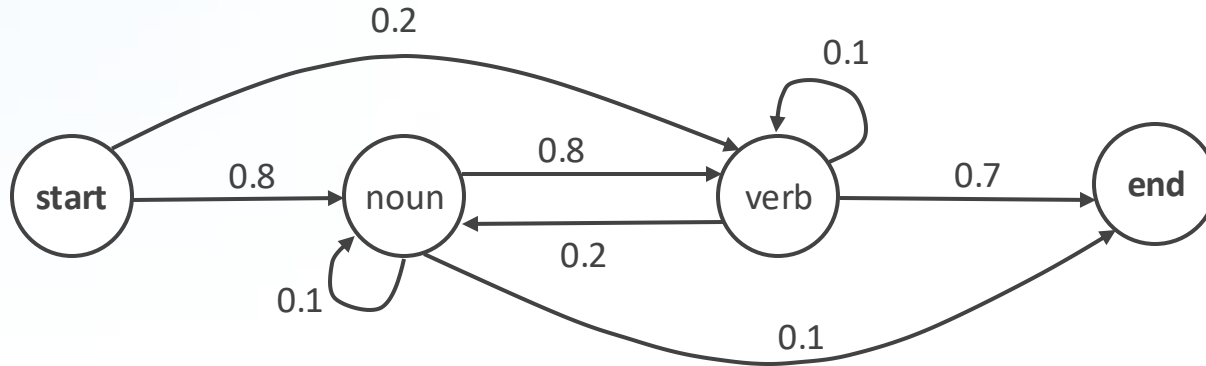
⋮


$$\frac{C(\text{verb sleep})}{C(\text{verb})} = \frac{5}{5+5} = \frac{5}{10}$$

Viterbi Probability

Based on [5]

Fish sleep.



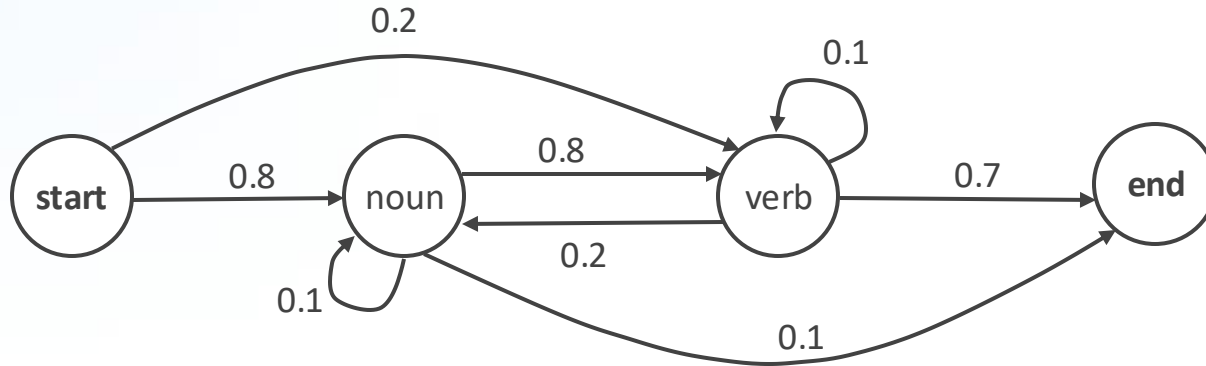
	0	1	2	3
start				
verb				
noun				
end				

Viterbi Probability

Based on [5]

Fish sleep.

Current token: <start>



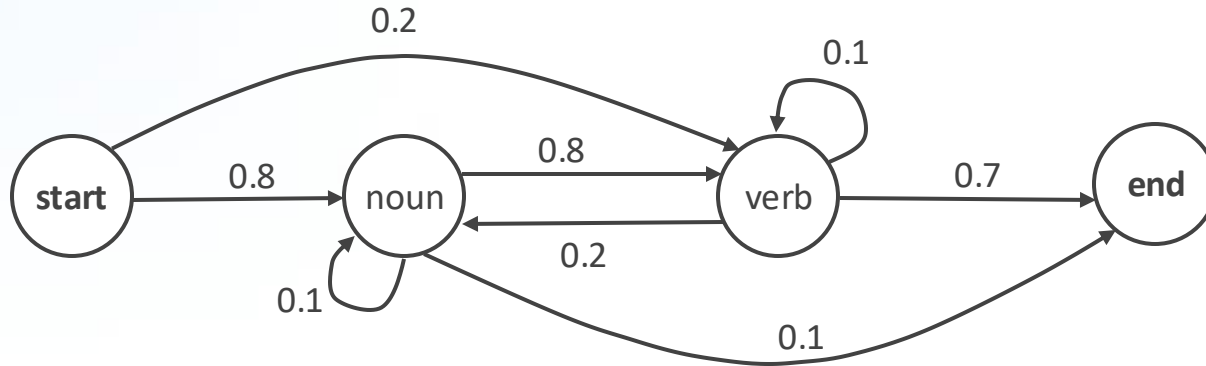
	0	1	2	3
start	1			
verb	0			
noun	0			
end	0			

Viterbi Probability

Based on [5]

Fish sleep.

Current token: fish



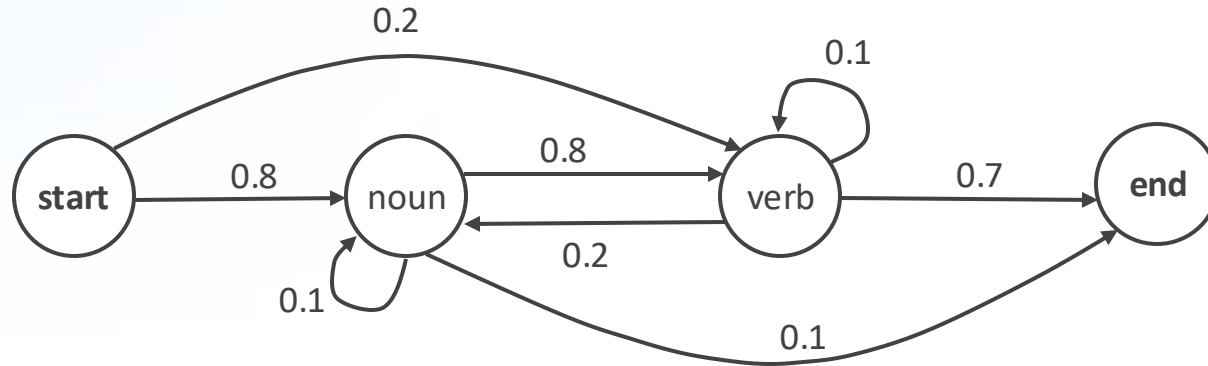
*transition * emission*

$$P(t_i|t_{i-1}) * P(w_i|t_i)$$

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$		
noun	0	$0.8 * 0.8 = 0.64$		
end	0			

Viterbi Probability

Based on [5]



Fish sleep.

Current token: sleep

If fish was a verb

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	$0.1 * 0.1 * 0.5 = 0.005$	
noun	0	$0.8 * 0.8 = 0.64$	$0.1 * 0.2 * 0.2 = 0.004$	
end	0			

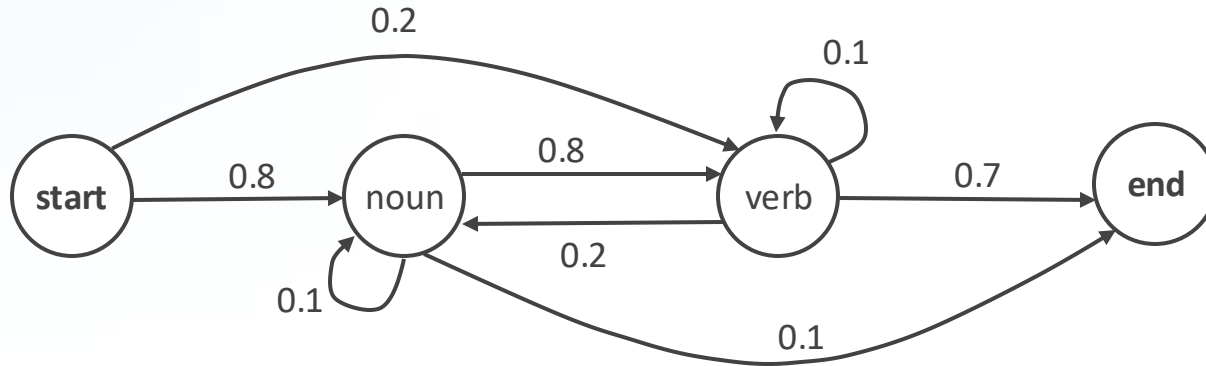
Viterbi Probability

Based on [5]

Fish sleep.

Current token: sleep

If fish was a noun



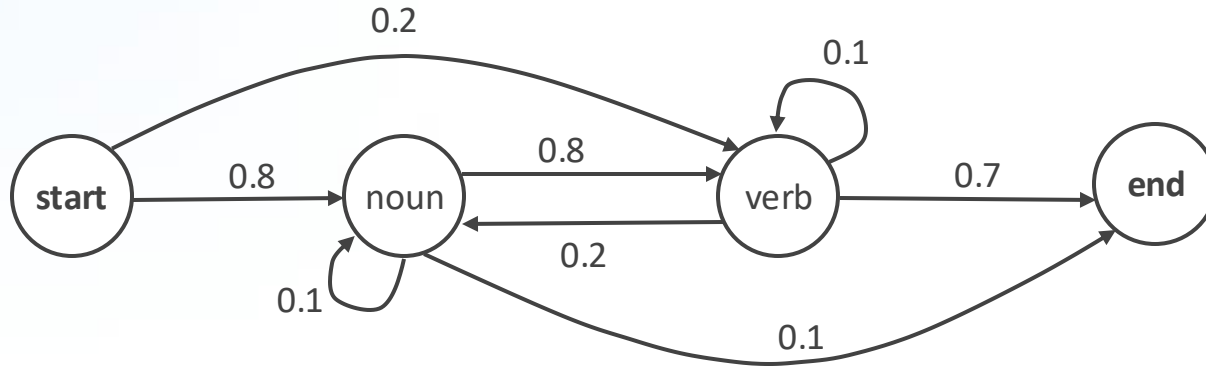
	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	$0.64 * 0.8 * 0.5 = 0.256$	
noun	0	$0.8 * 0.8 = 0.64$	$0.64 * 0.1 * 0.2 = 0.0128$	
end	0			

Viterbi Probability

Based on [5]

Fish sleep.

Current token: sleep



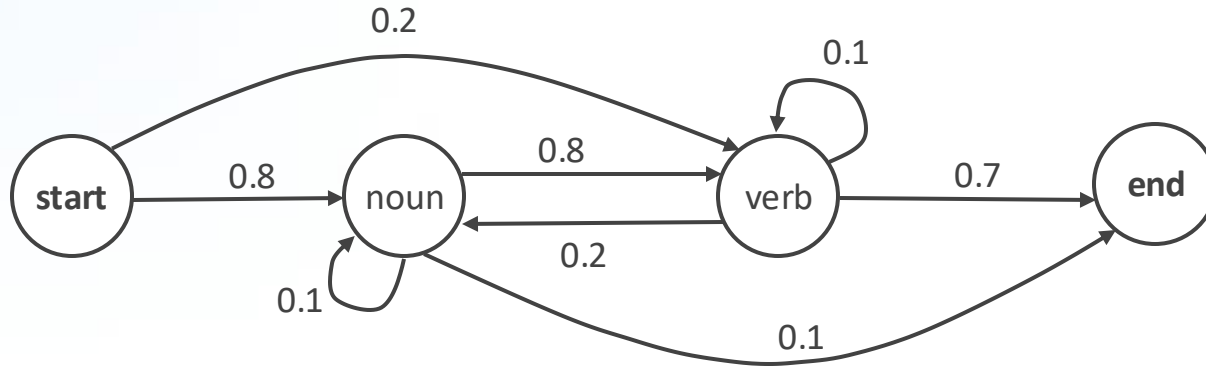
	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	0.005 0.256	
noun	0	$0.8 * 0.8 = 0.64$	0.004 0.0128	
end	0			

Viterbi Probability

Based on [5]

Fish sleep.

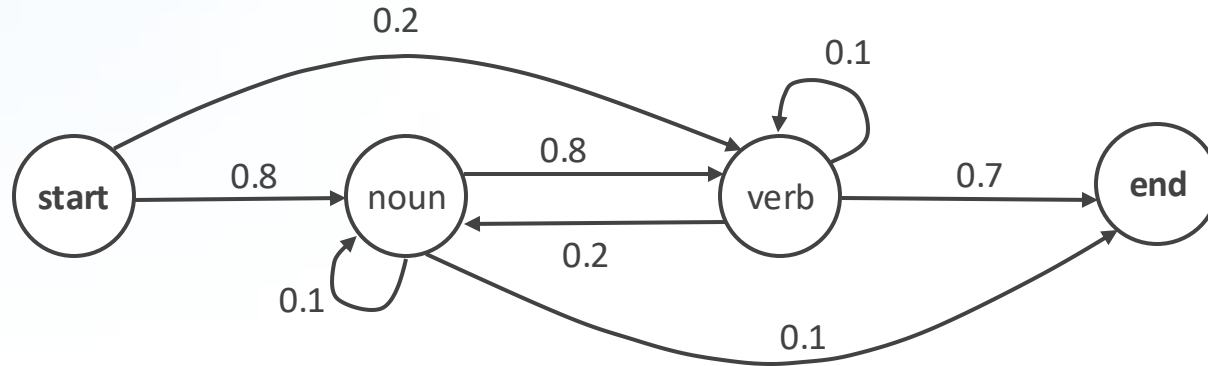
Current token: sleep



	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	0.005 0.256	
noun	0	$0.8 * 0.8 = 0.64$	0.004 0.0128	
end	0			

Viterbi Probability

Based on [5]



Fish sleep.

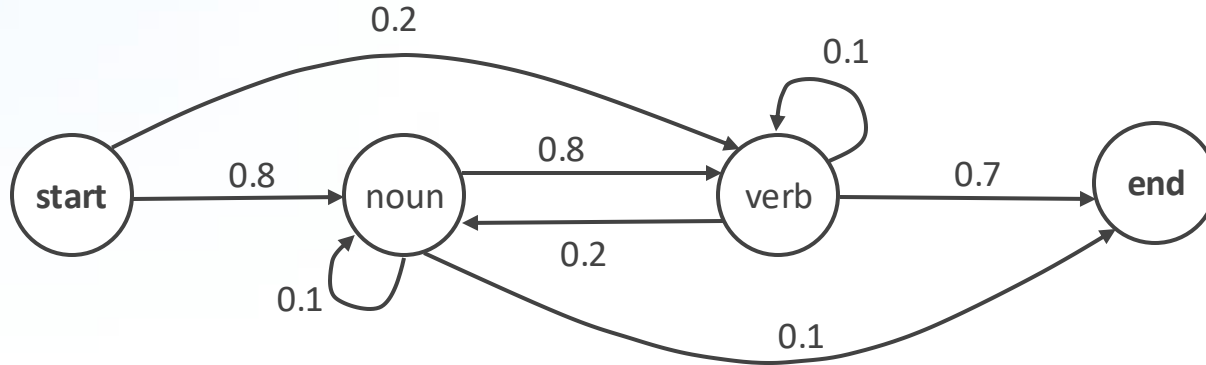
Current token: <end>

If sleep was a verb

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	0.005 0.256	
noun	0	$0.8 * 0.8 = 0.64$	0.004 0.0128	
end	0			$0.256 * 0.7 = 0.1792$

Viterbi Probability

Based on [5]



Fish sleep.

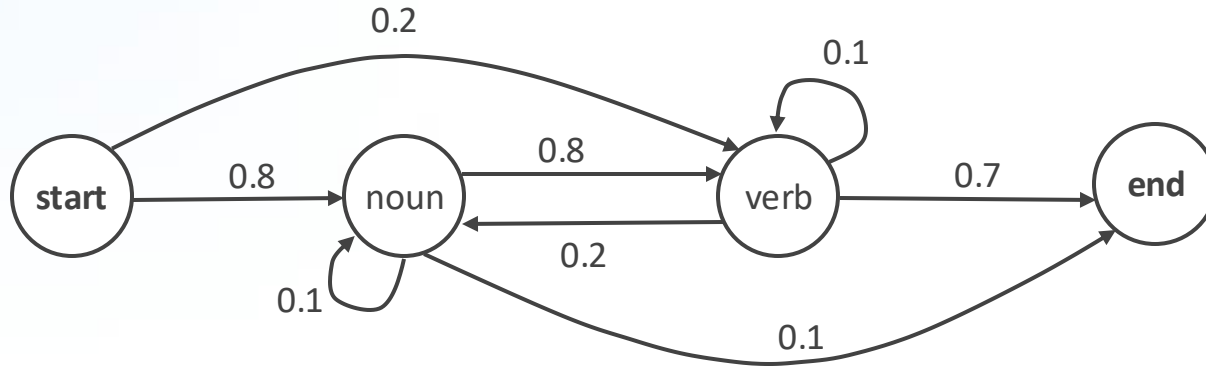
Current token: <end>

If sleep was a noun

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	0.005 0.256	
noun	0	$0.8 * 0.8 = 0.64$	0.004 0.0128	
end	0			$0.0128 * 0.1 = 0.00128$

Viterbi Probability

Based on [5]



Fish sleep.

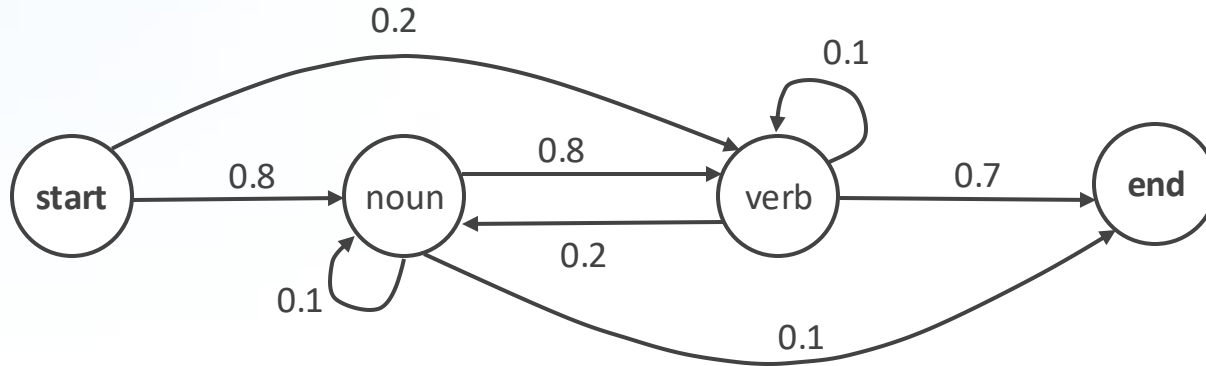
Current token: <end>

If sleep was a noun

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	$0.005 \quad 0.256$	
noun	0	$0.8 * 0.8 = 0.64$	$0.004 \quad 0.0128$	
end	0			$0.1792 \quad 0.00128$

Viterbi Probability

Based on [5]



Fish sleep.

Current token: <end>

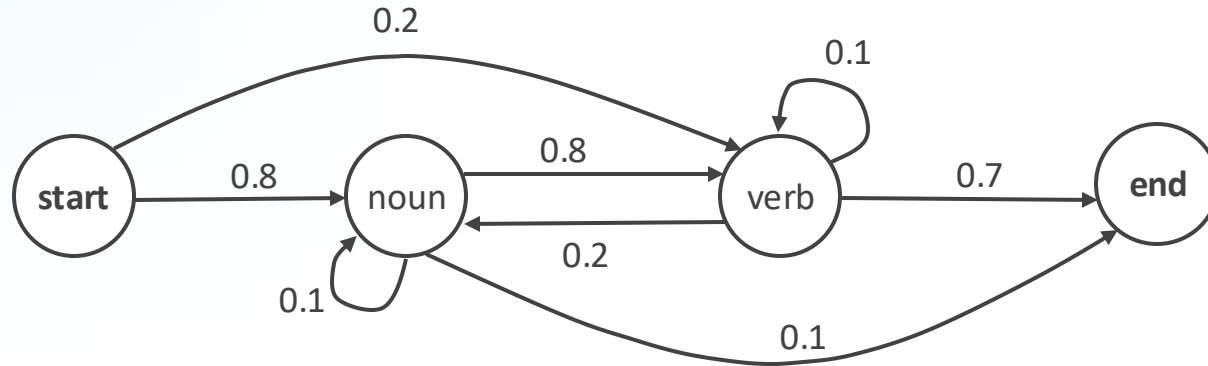
If sleep was a noun

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	$0.005 \quad 0.256$	
noun	0	$0.8 * 0.8 = 0.64$	$0.004 \quad 0.0128$	
end	0			$0.1792 \quad 0.00128$

Viterbi Probability

Based on [5]

Fish sleep.



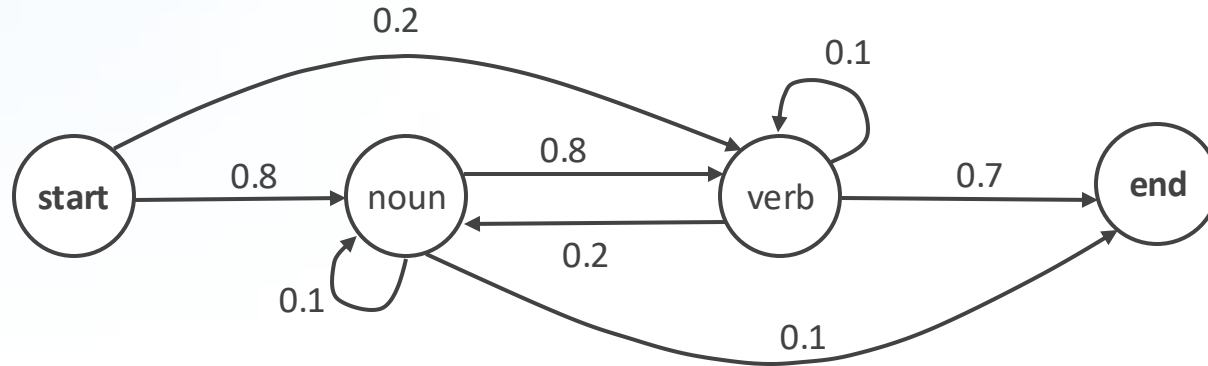
Decoding the result:
Sleep is a verb

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	0.005 0.256	
noun	0	$0.8 * 0.8 = 0.64$	0.004 0.0128	
end	0			0.1792 0.00128

Viterbi Probability

Based on [5]

Fish sleep.



Decoding the result:
Sleep is a verb
Fish is a noun

	0	1	2	3
start	1			
verb	0	$0.2 * 0.5 = 0.1$	0.005 0.256	
noun	0	$0.8 * 0.8 = 0.64$	0.004 0.0128	
end	0			0.1792 0.00128

Viterbi Complexity

The Viterbi algorithm complexity is $O(s^2n)$, where:

- s : number of states
- n : number of word/tokens in the input

Natural Language Processing

Parts of Speech

In this lecture:

- ☐ Part 1: POS Overview
- ☐ Part 2: POS Tagging (Viterbi algorithm)
- ☒ Part 3: **ALMA Morph Tagger**
- ☐ Part 4: Discussion



ALMA Morphology Tagger

- ❖ **Tasks:** Lemmatizer, POS tagger, and root tagger (Frequency-based)
 - ❖ **Open-Source**, part of [SinaTools](#)
 - ❖ **Performance:** lemmatization (90.48%), POS (93.8%), speed (32K tokens/sec)
 - ❖ **Benchmarking** the **accuracy** and **speed** of four tools
 - Alma, Madamira, CAMEL Tools, and Farasa
 - Using two corpora (ATB and Salma).
- ➔ **Alma outperformed all tools in all tasks.**

Mustafa Jarrar, Diyam Akra, Tymaa Hammouda: [ALMA: Fast Lemmatizer and POS Tagger for Arabic](#). In Proceedings of the 2024 AI in Computational Linguistics (ACLING 2024), Procedia Computer Science, Dubai. ELSEVIER.

ALMA Online Demo



<https://sina.birzeit.edu/alma/>

 SinaLab

NewsTeamResources

ALMA (ألمى)
Arabic Morphology Tagger

Lemmatizer, POS tagger, and root tagger.
Accuracy: POS (93.8%), lemmatization (90.48%), and speed (32K tokens/second). Outperformed all other tools (see [article](#)).

أسرع المحللات الصرفية العربية وأكثرها دقة، وهو محلل مفتوح المصدر. يتم تحديد المدخلة المعجمية وقسم الكلام والجذر لكل كلمة في النص.

Morph TaggerLemmatizerPOS Tagger

- Downloads

Download [SinaTools](#) (Morph Module), you can also access ALMA memory (morphological solution ordered by frequency) which is part of SinaTools.

Download [Qabas](#) lexicon that we used to build Alma.

+ Benchmarks

- Publications

Mustafa Jarrar, Diyam Akra, Tymaa Hammouda: [ALMA: Fast Lemmatizer and POS Tagger for Arabic](#). In Proceedings of the 2024 AI in Computational Linguistics (ACLING 2024), Procedia Computer Science, Dubai. ELSEVIER.

Tymaa Hammouda, Mustafa Jarrar, Mohammed Khalilia: [SinaTools: Open Source Toolkit for Arabic Natural Language Understanding](#). In Proceedings of the 2024 AI in Computational Linguistics (ACLING 2024), Procedia Computer Science, Dubai. ELSEVIER.

Alma Memory

- ❖ Alma retrieves only the most frequent solution regardless of the context.
- ❖ All solutions for all Arabic words are collected and ordered by frequency.
- ❖ Alma's memory is a lookup (8.7M word forms, 10.7M solutions):

```
[ {"Word Form": "بيت",
  "Solutions": [{"lemma": "بَيْت3", "pos": "NOUN", "root": "ب ي ت", "frequency": 596095},
                {"lemma": "بَيْت1", "pos": "NOUN", "root": "ب ي ت", "frequency": 140578},
                {"lemma": "بَيْت4", "pos": "NOUN", "root": "ب ي ت", "frequency": 9634},
                {"lemma": "بَيْت1", "pos": "PV", "root": "ب ي ت", "frequency": 3153} ] },
  {"Word Form": "بلال",
  "Solutions": [{"lemma": "بَلال1", "pos": "NOUN_PROP", "root": "ب ل ال", "frequency": null} ] },
  {"Word Form": "احلام",
  "Solutions": [{"lemma": "حُلُم1", "pos": "NOUN", "root": "ح ل م", "frequency": 212926},
                {"lemma": "حُلُم1", "pos": "NOUN", "root": "ح ل م", "frequency": 304} ] ] }
```

ALMA System

Alma algorithm steps:

1. **Exact-match-search:** Search for exact match of a word.
2. **Cleaning-search-again-loop:** Sequentially applies following normalization steps: (1) removing (ال) prefixes, (2) replacing final (ة) with (ه), (3) unifying Alef variants, (4) removing diacritics and (5) shadda. Then search again.
3. **Out-of-vocabulary-lemmatization:** retrieves a word with "_ 0" suffix.
4. **Out-of-vocabulary-root-tagging:** retrieves the wordform itself as root.

(Optional) Out-of-vocabulary-POS-tagging: predict a POS using a BERT model.



Alma

Part of SinaTools

(Open Source)



SinaTools

Open-source Python toolkit for Arabic Natural Understanding, allowing people to integrate it into their system workflow.

Python APIs, command lines, and online demos.

[Documentation](#)[GitHub](#)[Download](#)[MIT License](#)

Modules:

+ Morphology Tagging

+ Named Entity Recognition

+ Word Sense Disambiguation

+ Semantic Relatedness

+ Synonyms

+ Diacritic-Based Matching

+ Relation Extraction

+ Utilities

Morphology Tagging

SinaTools

Python API

```
from sinatools.morphology import morph_analyzer
morph_analyzer.analyze('ذهب الولد إلى المدرسة')
[{"token": "ذهب",
  "lemma": "ذَهِبَ",
  "lemma_id": "202001617",
  "root": "ذ ه ب",
  "pos": "فعل ماضي",
  "frequency": "82202"},
 {"token": "الولد",
  "lemma": "وَلَدٌ",
  "lemma_id": "202003092",
  "root": "و ل د",
  "pos": "اسم",
  "frequency": "19066"},
 {"token": "إلى",
  "lemma": "إِلَى",
  "lemma_id": "202000856",
  "root": "إ ل ي",
  "pos": "حرف جر",
  "frequency": "7367507"},
 {"token": "المدرسة",
  "lemma": "مَدْرَسَةٌ",
  "lemma_id": "202002620",
  "root": "د ر س",
  "pos": "اسم",
  "frequency": "145285"}]
```

Benchmarking Datasets

We used two datasets:

LDC's Arabic Treebank (ATB): Part 3 V3.2

- **339,710** tokens (202,377 nouns, 32,135 verbs, 105,198 particles).
- annotated with features like lemma, POS, segmentation, and gloss.
- It has 13,031 unique lemmas (10,602 nouns, 2,229 verbs, 214 particles).
- News wires between 2001 and 2011.

Salma Corpus:

- **34,253** tokens (19,030 nouns, 2,763 verbs, 12,460 particles)
- annotated with various features including lemmas and POS.
- It has 3,875 unique lemmas (2,904 nouns, 677 verbs, 294 particles).
- MSA media sources between 2021 and 2023.

Lemmatization Benchmarking

Benchmarking challenges

- Both corpora use SAMA lemmas
- Four lemmatizers produce different lemma spellings
- MADAMIRA retrieves SAMA lemmas
- Alma lemmas are mapped into SAMA lemmas
- Lemmas in CAMEL Tools very close to SAMA lemmas without digits.
- Farasa drops digits, diacritics, Shadda, and Hamza

Lemmatization Benchmarking

Five scenarios of lemmatization: (قَرَأْتُ بَيْتًا مِنَ الشَّعْرِ) (بَيْتٌ 1، بَيْتٌ 3، بَيْتٌ 4، بَيْتٌ 1)

→ True Lemmatization: exact matching. (بَيْتٌ 3)

→ Ambiguous Lemmatization: only digits are removing. (بَيْتٌ، بَيْتٌ)

→ Ambiguous undiacritized lemmatization: all diacritics except Shadda are removing. (بَيْت، بَيْت)

→ Loose lemmatization: Shadda is removing. (بَيْت)

→ Baggy lemmatization: all Hamza forms are normalizing. (أَنْأ، إِنْأ) → (أَنْأ، إِنْأ)

Lemmatization Benchmarking

Experiment	ATB	SALMA
Exp1: True Lemmatization (Exact Match)		
ALMA	87.82%	90.48%
MADAMIRA	84.18%	85.53%
CAMeL Tools	-	-
Farasa	-	-
Exp2: Ambiguous lemmatization (without Numbers)		
ALMA	91.50%	91.17%
MADAMIRA	88.35%	86.60%
CAMeL Tools	80.88%	81.73%
Farasa	-	-
Exp3: Ambiguous undiacritized lemmatization		
ALMA	93.25%	93.40%
MADAMIRA	96.08%	93.24%
CAMeL Tools	90.65%	90.83%
Farasa	68.92%	70.78%
Exp4: Loose Lemmatization (striped)		
ALMA	94.42%	94.97%
MADAMIRA	96.73%	94.48%
CAMeL Tools	93.64%	93.29%
Farasa	86.73%	91.28%
Exp5: Baggy lemmatization (striped and normalized)		
ALMA	95.01%	95.88%
MADAMIRA	96.94%	94.55%
CAMeL Tools	94.81%	94.73%
Farasa	94.86%	95.33%

Alma Lemmatization Errors

Types of Errors :

- *Ambiguous Lemmatization*: Alma retrieves (سِيَّاسَة 1) while (سِيَّاسَة 2) is the correct one.
- *Wrong Lemmatization*: Alma retrieves the lemma (هُمْ) for word (فهم).
- *Out of vocabulary Lemmatization*: word not found in Alma memory (ترامب)

Error Category	ATB				Salma			
	Noun	Verb	Particle	All	Noun	Verb	Particle	All
Ambiguous	39.2% ¹³⁵⁰¹	9.7% ³³⁴⁷	11.9% ⁴⁰⁷⁷	60.7%²⁰⁹²⁵	36% ⁹⁷⁵	7.0% ¹⁹²	18.0% ⁴⁹⁰	61.2%¹⁶⁵⁷
Wrong	23.8% ⁸²⁰⁸	6.5% ²²³⁶	8.9% ³⁰⁶³	39.2%¹³⁴⁹⁵	20.6% ⁵⁵⁷	11.7% ³¹⁶	6.4% ¹⁷³	38.7%¹⁰⁴⁶
OOV	0.10 % ³⁵	0.009% ³	0.02% ⁸	0.13%⁴⁶	0.18% ⁵	0.04% ¹	-	0.22%⁶

POS Tagging Benchmarking

- ❖ Alma and MADAMIRA use the SAMA tagset (39 tags).
- ❖ CAMEL Tools uses these 39 tags but with different naming.
- ❖ Farasa uses a tagset of 18 tags.
- ❖ Three types of POS tagging scenario:
 - **True POS tagging:** used the full 39 tags used in the ATB
 - **Simplified POS tagging:** reduced the 39 tags into 13 simplified tags.
 - **POS Category tagging:** reduced the 39 tags into 3 tags/categories.

POS Tagging Benchmarking

	ATB		
	True POS	POS Simplified	POS Category
<i>Alma</i>	92.7%	93.1%	97.5%
<i>Alma</i> +BERT	98.2%	98.6%	99.5%
<i>Alma</i> +BERT(OOV)	92.7%	93.1%	97.6%
BERT	98.1%	98.6%	99.5%
MADAMIRA	82.3%	90.8%	97.5%
CAMeL Tools	82.7%	91.9%	98.6%
Farasa	62.4%	84.9%	94.3%

ALMA POS Tagging Errors

Types of Errors:

- General POS: (آليات) tagged as **ADJ** while its **NOUN**.
- Wrong POS: (أحب) tagged as **IV** while its **ADJ_COMP**.
- OOV: Alma fails to retrieve a POS for a word.

	ATB			
	Noun	Verb	Particle	All
General POS	48.4% ¹⁰⁰¹²	4.5% ⁹³⁸	6.2% ¹²⁸¹	58.9%¹²¹⁷²
Wrong POS	14.2% ²⁹³³	14.9% ³⁰⁷⁵	11.8% ²⁴²⁸	40.9%⁸⁴⁴⁴
OOV	0.22% ⁴⁷	0.03% ⁸	0.02% ⁵	0.29%⁶⁰

Speed Benchmarking

- ❖ Speed of the four tools is evaluated:
 - **same machine** (24 CPU, 47G Memory, CentOS, size 1.3T)
 - **same setup** (reading input from a file and writing output to a file).
 - the **default settings** in all tools for lemmatization and POS tagging are used without changing any configuration.
- ❖ The experiment was repeated **six times**:
 - excluding the first run (initial model loading time is not considered).
 - the reading and writing of input/output files are considered

Speed Benchmarking

- ❖ Averages and standard deviations.
- ❖ Alma is significantly faster (34K tokens/second).
- ❖ This speed is attributed to ALMA's architecture, which uses a Python dictionary for fast memory access and processing.

Experiment	ATB (seconds)	SALMA (seconds)	Speed (tokens per second)
ALMA	10 ± 1	$1 \pm .11$	33,997
MADAMIRA	1710 ± 99	370 ± 49	180
CAMeL Tools	$14,398 \pm 1234$	677 ± 28	25
Farasa	35 ± 2	14 ± 2	7632

Natural Language Processing

Parts of Speech

In this lecture:

- ☐ Part 1: POS Overview
- ☐ Part 2: POS Tagging (Viterbi algorithm)
- ☐ Part 3: ALMA Morph Tagger
- ☒ Part 4: **Discussion**



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